



Skin Disease Diagnosis Report

1. Report Information

This report has been automatically generated by the Skin Disease Classification & Diagnosis Platform using a trained deep learning model. It summarizes the prediction results for the uploaded skin lesion image and provides supporting information to help interpret the model's decision.

- Report ID
- Report Generation Date & Time
- AI Model Used
- Model Version

2. Uploaded Image Details

The uploaded skin lesion image was preprocessed before prediction. Preprocessing included resizing the image to the required input dimensions, normalizing pixel values, and preparing the image in a format compatible with the trained neural network. These steps ensure consistent and reliable predictions.

- Uploaded Skin Lesion Image
- Image Filename
- Image Resolution
- Image Format

3. Prediction Summary

The AI model analyzed the uploaded image and identified the most probable skin disease category. Along with the predicted class, the model provides a confidence score indicating how certain it is about the prediction. A higher confidence score generally suggests greater certainty in the model's classification.

- Predicted Disease
- Confidence Score
- Prediction Status (Successful)

4. Class Probability Distribution

The model calculates the probability of the uploaded image belonging to each of the seven disease categories. Instead of considering only the final prediction, these probabilities provide a complete overview of the model's confidence across all possible classes, helping users understand alternative predictions.

Disease	Probability
Melanoma	XX.XX%
Melanocytic Nevus	XX.XX%
Basal Cell Carcinoma	XX.XX%
Benign Keratosis	XX.XX%
Actinic Keratosis	XX.XX%
Dermatofibroma	XX.XX%
Vascular Lesion	XX.XX%

5. Explainable AI (Grad-CAM)

To improve transparency, the report includes a Grad-CAM (Gradient-weighted Class Activation Mapping) visualization. The heatmap highlights the regions of the skin lesion that had the greatest influence on the model's prediction, allowing users to better understand how the AI reached its decision.

- Original Uploaded Image
- Grad-CAM Heatmap
- Visual Explanation of the Predicted Region

6. Diagnosis Summary

Based on the analysis, the model predicts the most likely skin disease category for the uploaded image. The diagnosis summary combines the predicted class, confidence score, probability distribution, and Grad-CAM explanation to provide a concise overview of the prediction outcome.

- Predicted Disease
- Confidence Level

- Interpretation of the Prediction
- Model Decision Summary

7. AI Model Information

This prediction was generated using a Convolutional Neural Network (CNN) trained on the HAM10000 skin lesion dataset. The model was developed using TensorFlow/Keras and optimized to classify seven different categories of skin diseases from dermoscopic images.

- Model Name
- Input Image Size
- Number of Output Classes
- Framework (TensorFlow/Keras)

8. Recommendation

The prediction generated by the AI model should be used as a decision-support tool rather than a definitive diagnosis. If the lesion appears suspicious, changes in appearance over time, or is associated with symptoms such as pain, itching, or bleeding, consultation with a qualified dermatologist is strongly recommended for clinical evaluation and appropriate treatment.

- This prediction is generated by an AI model and is intended for educational and research purposes.
- The result should not be considered a final medical diagnosis.
- Please consult a qualified dermatologist or healthcare professional for clinical evaluation and treatment.